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Adductor longus mechanics during the maximal effort soccer kick

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Abstract
Groin pain is a common cause of athletic disability and often involves the adductor longus. A common complaint of patients with groin problems is pain while preparing to kick the ball. The purpose of this study was to examine muscle length and activation of the adductor longus while kicking a soccer ball. Three-dimensional joint positions and muscle activation were obtained from 15 National Collegiate Athletic Association (NCAA) Division 1 male soccer players during maximal effort kicks. Musculoskeletal modeling techniques incorporating joint position and muscle attachments were used to estimate adductor longus length from the beginning of the kicking leg's swing phase until ball strike. The maximum rate of stretch of the adductor longus (22.3 ± 5.3 cm/s) and maximum hip extension (23.3 ± 8.8°) occurred near 40% of swing phase. Activation of the adductor longus occurred between 10% and 50% of the swing phase. Adductor longus maximum length occurred at 65% of the swing phase. Maximum hip abduction (25.3 ± 5.4°) occurred at 80% of swing phase. The adductor longus appears to be at risk of strain injury during its transition from hip extension to hip flexion. This knowledge could be applied to muscle injury prevention and rehabilitation programs to aid with treatment of adductor longus related groin pain.

Keywords: *Adductor longus, Electromyography (EMG), groin injuries, muscle modeling*

Introduction

Soccer is the most popular sport in the world with over 265 million registered players worldwide (FIFA). The most common injury locations reported during four World Cup tournaments were the thigh (23%–24%), ankle (18%–19%), and knee (15%–17%) (Giza et al., 2003). Groin injuries have been reported to account for between 5% and 13% of all soccer injuries and can lead to substantial time loss from both practice and competition (Gilmore, 1998; Lynch and Renstrom, 1999; Nicholas and Tyler, 2002; Wong and Hong, 2005). Potential risk factors for groin injuries are thought to be previous strains, direct trauma, overuse, and muscular imbalance between the abdominal muscles and adductors (Gilmore, 1998; Akita et al., 1999).

Common types of chronic groin injuries include dysfunction of the adductor longus myotendinous unit, impingement of the inguinal nerves, posterior inguinal wall deficiency

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(sportsman's hernias), osteitis pubis, and abdominal strains near the insertion on the pelvic floor (Taylor et al., 1991; Bradshaw and McCrory, 1997; Akita et al., 1999; Lynch and Renstrom, 1999; Anderson et al., 2001; Morelli and Smith, 2001; Cowan et al., 2004; Robinson et al., 2004). Diagnosis and treatment are often difficult because the causes of chronic groin pain are poorly understood and the symptoms often overlap disparate injuries (Ekberg et al., 1988; Gilmore, 1998; Lynch and Renstrom, 1999; Susmallian et al., 2004; Cetin et al., 2004; Slavotinek et al., 2005).

To treat the various causes of athletic groin pain most effectively, it is important to have a clear understanding of the mechanisms of injury. Athletic groin pain has been studied extensively and one potential mechanism that has been mentioned is repetitive stressing of the hip adductors and abdominal muscles during a kicking motion (Taylor et al., 1991; Ekberg et al., 1988; Akita et al., 1999; Meyers et al., 2000; Albers et al., 2001; Anderson et al., 2001; Verrall et al., 2001; Cetin et al., 2004; Robinson et al., 2004; Slavotinek et al., 2005; Verrall et al., 2005; Verrall et al., 2007) even though this mechanism for injury has never been explicitly studied or defined. The mechanics of kicking a soccer ball have been studied with specific emphasis on comparing the side foot and instep kicks (Levanon and Dapena, 1998; Nunome et al., 2002), investigating differences in kicking mechanics between the preferred and non-preferred leg, (Dorge et al., 2002) and describing the movements of the thigh, lower leg, and knee (Kellis et al., 2004). These, and other related biomechanical studies, focused on creating a biomechanical model of the kicking motion or optimizing foot speed, resultant ball speed, and kicking efficiency (Barfield, 1998; Lees and Nolan, 1998; Levanon and Dapena, 1998; Dorge et al., 2002; Nunome et al., 2002).

While kicking mechanics have been examined, there is little information examining potential injury mechanisms associated with kicking. Only one study addressed the role of the adductor longus during kicking, reporting that the adductors were more active during an instep kick than during a side-foot kick (Brophy et al., 2007). We are unaware of any studies examining how adductor longus activation, lower extremity kinematics, and adductor longus muscle length during kicking could be related to possible mechanisms of injury to the adductor muscles.

Previous hamstring muscle injury literature has indicated that a muscle is at greatest risk of injury when it is eccentrically contracting (Dornan, 1971; Glick, 1980; Zarins and Ciullo, 1983; Stanton and Purdam, 1989; Stauber, 1989; Garrett, 1990; 1996; Kellis and Baltzopoulos, 1995; Kuljala et al., 1997). Our purpose was to investigate the three-dimensional position of the hip and the adductor longus length and activation during the swing phase of a maximal instep soccer kick. Based on the current literature, we expected to see that eccentric activation of the adductor longus would occur during the middle portion of the swing phase when the kicking leg is extending and abducting. This would mean that the adductor longus may be at greatest risk of injury during the middle portion of the swing phase as the leg reached maximum extension and abduction.

Methods

Fifteen male, NCAA Division I soccer players (mean age 20.5 ± 1.4 years, mean height 1.8 ± .07 m, mean weight 77.1 ± 9.3 kg) volunteered to be subjects. Prior to testing, each subject gave written informed consent that was approved by the medical center's institutional review board. Prior to testing, the kick was demonstrated for the subject. Treadmill jogging and kicking warm-up time as well as practice trials were allowed if the subject desired. Kinematic data were collected using an eight-camera motion capture system collecting at 240 Hz (Motion Analysis, Santa Rosa, CA). Each subject had 31 retro-reflective markers

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(plus one offset tracking marker) placed at the following anatomic landmarks: the joint between the fourth and fifth lumbar vertebrae as well as bilaterally on the acromioclavicular joint of the shoulder, elbow, wrist, anterior superior iliac spine, greater trochanter, thigh, medial and lateral joint line of the knee, shank, medial and lateral malleolus, medial, lateral and middle heel, and the head of the fifth metatarsal. Next, a pair of electromyography surface electrodes (Noraxon, Scottsdale, AZ) were placed on the adductor longus of the kicking leg (right leg) (Konrad, 2005). Electrode placement was verified with manual muscle testing in order to determine that that EMG electrodes were accurately placed on the adductor longus. The EMG data were collected at 1200 Hz and were time synchronized with the kinematic data during collection using EvaRT v5.0 (Motion Analysis Corporation, Santa Rosa, CA).

All subjects were currently active at least three times per week, had no previous history of chronic groin pain, and no lower extremity or abdominal injuries in the previous six months. The ball used was an official size and weight Nike Geo (Nike Inc., Beaverton, OR). A piece of retro-reflective tape was placed on top of the ball to aid in the identification of ball contact during data reduction. Subjects wore spandex shorts and appropriately sized Nike Air Pegasus running shoes to minimize potential variability due to footwear conditions. Each subject was asked to stand in the middle of the capture volume for a static calibration trial. Following the static trial, the medial knee and ankle markers were removed for the remainder of testing. Subjects were then asked to perform seven acceptable maximal in-step soccer kicks, similar to a direct shot, into a large net located approximately 2 m from the ball (Figure 1). The subjects were asked to use the leg they would use for a penalty kick. In all cases, this was the right leg. The data was collected from the start of the approach until the end of follow-through when the subject returned to a standing position. An acceptable trial was defined as a kick where all of the motion capture and EMG data were collected properly and the kick was accurate and powerful. An average of nine attempts was needed to obtain the seven acceptable trials.

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Data reduction

Motion capture data were processed and smoothed using a recursive fourth order Butterworth low-pass filter with a cut-off frequency of 6 Hz. The use of a 6 Hz cut-off frequency could result in distortion of the kinematic data, especially for distal segments, near ball impact. However, this study focused on the portion of the swing phase near peak hip extension and did not examine the distal segment, therefore the cut-off frequency of 6 Hz was appropriate. The swing phase for data reduction was defined as the time from toe off of the kicking foot until ball contact with the kicking foot (Figure 2). All of the EMG and kinematic data were normalized to the swing phase with toe off being 0% and ball contact being 100%. In addition, heel strike of the contralateral foot (plant leg) was recorded as a percentage of the swing phase. Using OrthoTrak v6.3.4c (Motion Analysis Corporation, Santa Rosa, CA), the maximum flexion/extension, abduction/adduction, and internal/external rotational values were identified for the kicking hip and the peak sagittal plane angles were identified for the pelvis and knee. The time at which each of these peaks occurred was also recorded as a percentage of the swing phase.

EMG data were output to MyoResearch XP v1.06.53 (Noraxon, Scottsdale, AZ) for data reduction. EMG data were first rectified and then the data was filtered using a recursive bandpass filter with cut-off frequencies of 450 Hz and 8 Hz. The data were then normalized to a previously determined peak activation. The peak activation of the adductor longus was defined as the maximum activation in any trial. This single value was then used to normalize the activation for all trials. The band-pass filtered rectified EMG data were filtered through

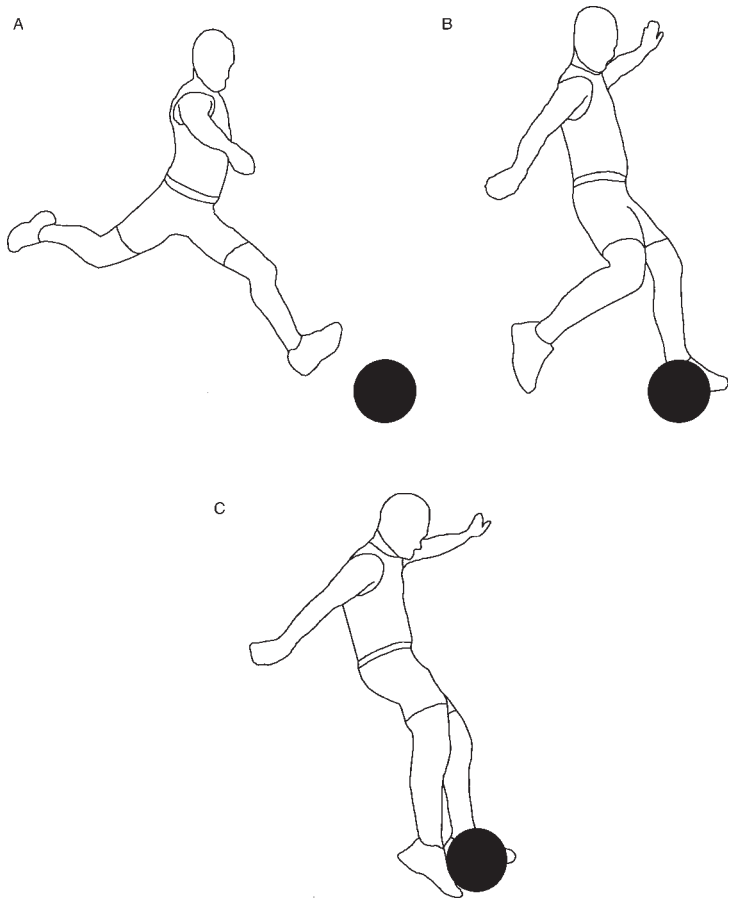

Figure 1. Representation of the soccer kicking progression from the peak extension of the kicking hip to ball contact during data collection.

a low-pass digital filter again at a cut-off frequency of 20 Hz to obtain linear envelop EMG data. Peak values, reported in units of peak activation, and time to peak values were reported as a percentage of the swing phase.

Musculoskeletal model

A three-dimensional musculoskeletal model was used to estimate adductor longus muscle length in different hip positions. The model was based on a bilateral model developed by Carhart (2003), that Lewis et al. (2009) simplified to include only four segments: the pelvis, thigh, shank and foot of the kicking (right) leg. The model contained six rotational degrees of freedom to represent the primary motions at the hip (adduction/abduction, flexion/extension, and internal/external rotation), knee (flexion/extension) and ankle (dorsiflexion/plantar flexion, inversion/eversion).

Musculoskeletal parameters, including muscle path, were adapted from work by Delp et al. (1990) for the adductor longus. For this study, the length of the adductor longus muscle was calculated as the distance between the origin point on the pelvis (the medial portion of the superior pubic ramus) and the insertion point on the linea aspera of the femur given the three-dimensional hip angle for each frame of data. The rate of change in length was also

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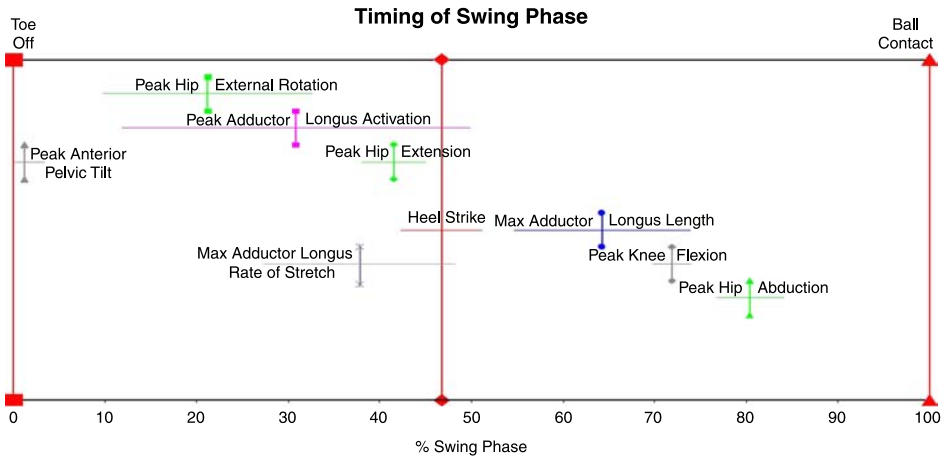

Figure 2. Timing of dependent variables in the swing phase of a maximal effort instep soccer kick.

calculated as the change of muscle length between frames and is expressed in cm/frame. To investigate how changes in flexion, abduction and external rotation individually contributed to muscle length, we decomposed the length change into the change attributable to each angle during the soccer kick. For example, we calculated the muscle length at time t of the kicking motion using the flexion, abduction and rotation angles. We then calculated the muscle length advancing only the angle in the sagittal plane (flexion/extension) to time $t+1$ while using the abduction and rotation angles from time t during the kick. The difference between the two calculated lengths was the change in muscle length due to flexion alone. This was calculated for each angle throughout the entire swing phase.

Data analysis

The first three acceptable trials of the seven trials collected were averaged for kinematic, EMG and muscle length data for each subject. This averaged data was then used to obtain summary graphs of the three-dimensional hip rotation angles, adductor longus length, and the maximum EMG activation of the adductor longus across all of the subjects throughout the swing phase (Figure 3 A–E). In addition, the percent of the swing phase at which these peak values occurred were also averaged across subjects (Table I).

Results

For the current study, toe off of the kicking leg was defined as the starting point (0%) of the swing phase of the kick. Peak anterior tilt of the pelvis occurred first followed by peak external rotation of the kicking hip (Table I). Maximum activation of the adductor longus occurred next followed directly by the maximum rate of lengthening of the adductor longus (Table I). Following the maximum rate of lengthening of the adductor longus peak hip extension of the kicking leg occurred shortly followed by contralateral foot strike (plant foot) (Table I). The adductor longus then reached maximum length followed by peak knee flexion of the kicking leg (Table I). Finally, peak abduction of the kicking hip occurred shortly before ball contact (Table I, Figure 2).

Peak hip extension angle, maximum rate of adductor longus length change, and maximum activation of the adductor longus all occurred between 30% and 45% of the swing phase.

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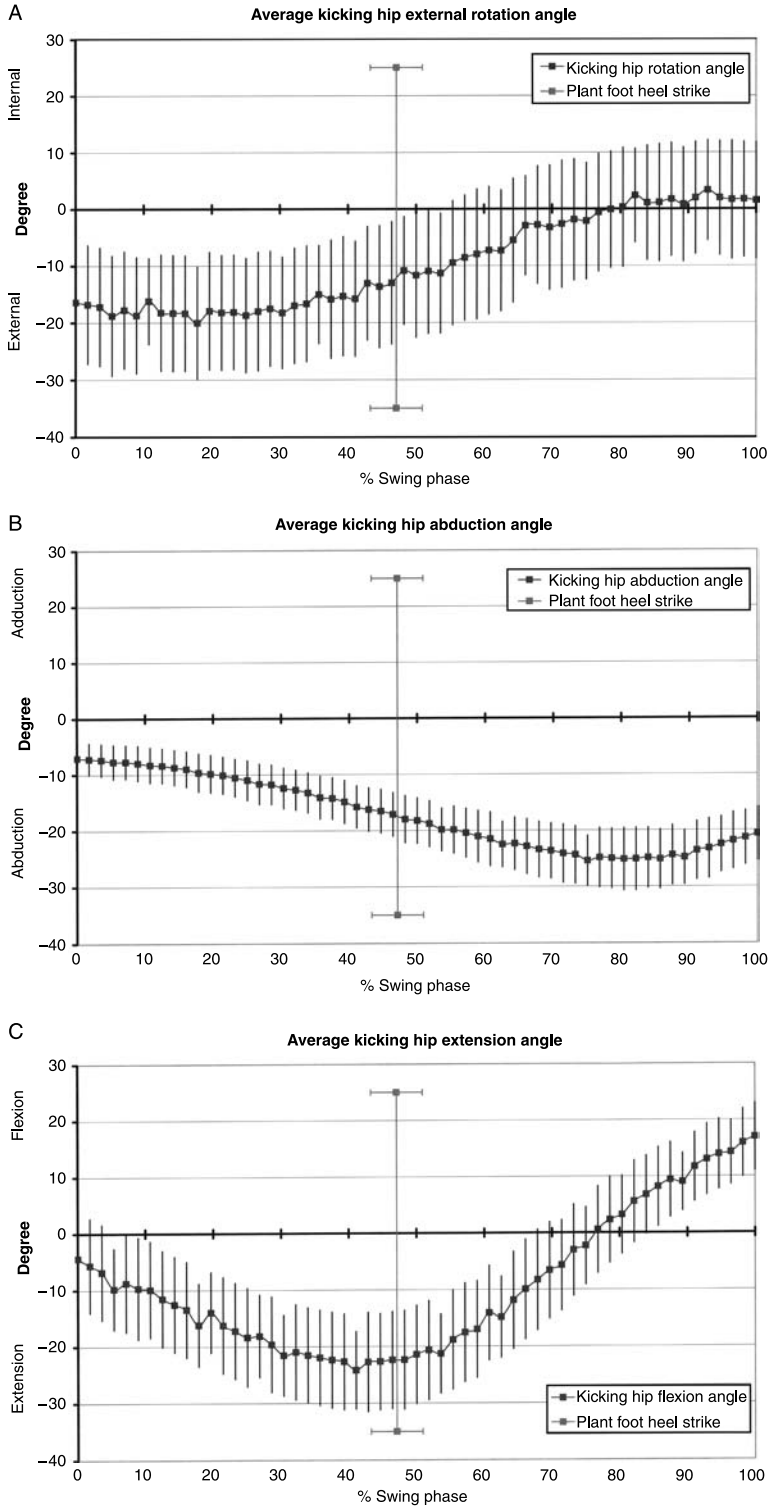

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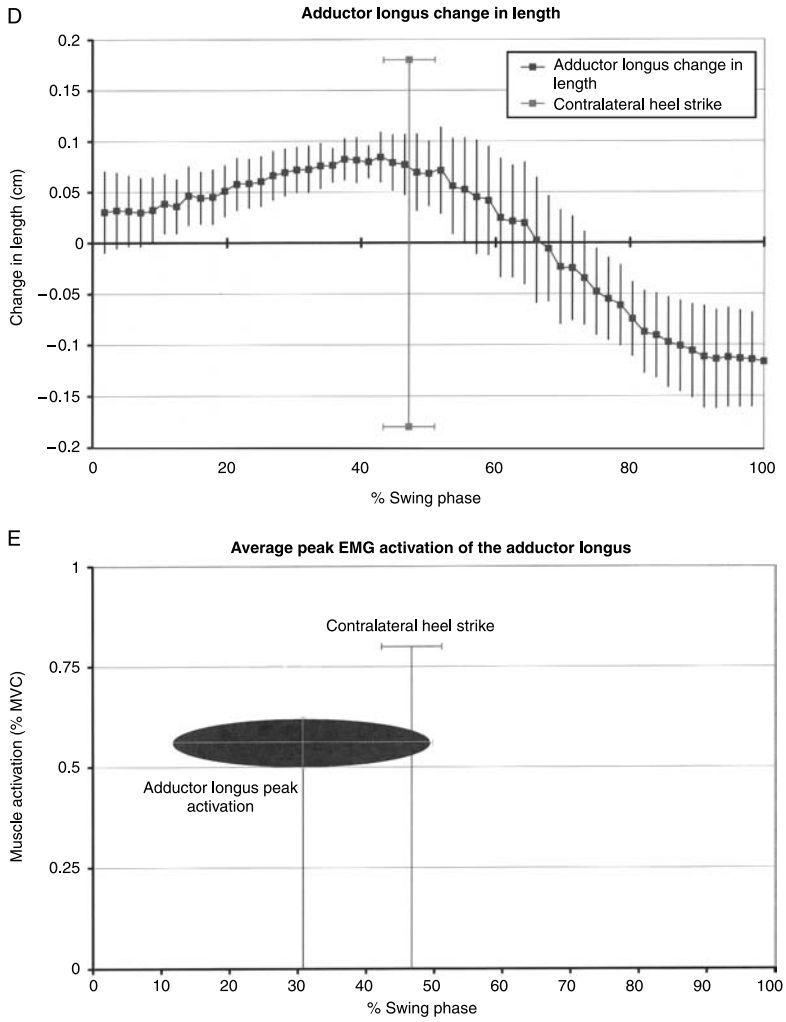

Figure 3. (A–C) Average kicking hip angles, (D) adductor longus length and (E) peak EMG muscle activation across subjects.

The kicking hip was abducted throughout the swing phase and was externally rotated for the majority of the swing phase until just prior to ball contact (Figure 3). Early in the swing phase, the lengthening of the adductor longus is mostly the result of increased hip extension and abduction (Figure 4). Shortly after the adductor longus reaches its maximum rate of lengthening, the kicking leg reaches peak hip extension. After this point in the kicking cycle, it appears that the decrease in the rate of lengthening of the adductor longus is mostly due to the change in the hip flexion angle (Figure 4).

Discussion and implications

Muscle strain injuries are thought to occur during eccentric contraction and are usually evident at the myotendinous junction (Dornan, 1971; Glick, 1980; Zarins and Ciullo, 1983; Stanton and Purdam, 1989; Stauber, 1989; Garrett, 1990; 1996; Kellis and Baltzopoulos, 1995; Kuljala et al., 1997). The purpose of this study was to determine when the adductor

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Table I. Kinematics, EMG activation and timing during the maximal effort soccer kick.

	Percent swing ± *s*	Peak value ± *s*
Toe off kicking foot	**0 ± 0%**	
Maximum anterior pelvic tilt	1.2 ± 2.2%	28.6 ± 6.6°
Maximum kicking hip external rotation	21.2 ± 11.5%	19 ± 9.8°
Maximum activation of the kicking hip adductor longus	30.8 ± 19%	0.56 ± 0.06 PA
Maximum rate of stretch of the kicking hip adductor longus	37.8 ± 10.4%	22.3 ± 5.3 cm/s
Maximum extension of the kicking hip	41.5 ± 3.5%	23.3 ± 8.8°
Heel strike of the plant foot	**46.7 ± 4.4%**	
Maximum length of the kicking leg adductor longus	64.2 ± 9.6%	
Maximum flexion of the kicking knee	71.9 ± 2.1%	90.3 ± 9.6°
Maximum abduction of the kicking hip	80.4 ± 3.7%	25.3 ± 5.4°
Ball contact kicking foot	**100 ± 0%**	

longus is at the highest risk of strain injury during the swing phase of the maximal effort soccer kick. The adductor longus was active throughout the swing phase, but the maximum activation and maximum rate of stretch occurred immediately prior to peak extension of the kicking hip (41.5 ± 3.5%) (Figure 2). The adductor longus could be at risk of strain injury until the muscle reaches its maximum length at 64.2 ± 9.6% of the swing phase, which is immediately prior to peak abduction of the kicking hip (Figure 2).

The timing of adductor longus maximum activation, maximum rate of stretch, and hip extension implies an eccentric contraction of the adductor longus during which the adductor longus could be activated to help control hip extension (Figure 2). During the maximal effort soccer kick, the adductor longus appears to be at risk of injury around the time of peak hip extension. Shifting the muscle activation data (van Ingen Schenau et al., 1992) to account for electromechanical delay (~40 ms) (Winter and Brookes, 1991) places peak activation around 45% of swing, immediately following peak hip extension. After this point, the adductor longus continued to lengthen but the rate of stretch became more gradual until maximum adductor longus length was reached at 64.2 ± 9.6% of the swing

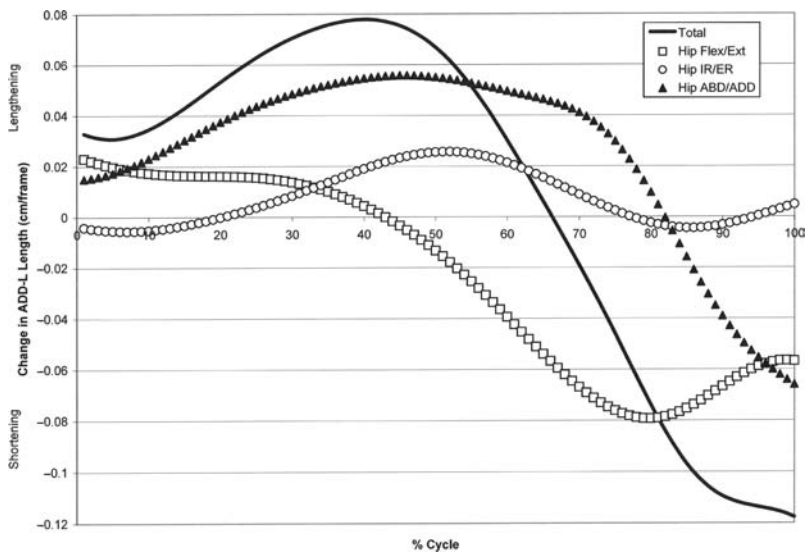

Figure 4. Total change in adductor longus length and change in adductor longus length due to hip flexion/extension, internal/external rotation, and abduction/adduction during a maximal soccer kick.

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phase. This was prior to peak abduction of the kicking hip, which occurred at approximately 80% of the swing phase (Figure 4). During this portion of the swing phase, the adductor longus was active, although not at its peak value. This implies a less powerful continuation of the eccentric contraction until maximum length was reached. According to the mechanism of strain injury previously discussed, the adductor longus appears to remain at risk for injury throughout the swing phase from toe off until maximum adductor longus length is achieved around 65% of the swing phase.

The timing and peak values of external rotation, extension, and abduction of the kicking hip and knee flexion angles observed in the current study are consistent with those of previous studies (Levanon and Dapena, 1998; Nunome et al., 2002; Kellis and Katis, 2007). The peak values for activation obtained in the current study are similar in magnitude to values reported for other muscles in previous studies examining the same kicking motion (DeProft et al., 1988; Dorge et al., 1999; Manolopoulos et al., 2006; Brophy et al., 2007; Kellis and Katis, 2007). We are unaware of any biomechanical studies of kicking that have used musculoskeletal modeling to describe either the length or change in length of the adductor longus during the maximal effort soccer kick in an attempt to relate kinematics to the injury mechanism for adductor-related groin pain.

Laboratory studies of sport skills have several inherent limitations that reduce external validity. For example, the ball is kicked off the floor instead of being kicked off grass. Running shoes are used instead of cleated shoes and no laboratory setting can mimic training or competition, however, our subjects were instructed to complete the task with maximal effort. The entire skill also may be impacted when wearing the equipment needed to access muscle activation. This may be reflected in the timing of EMG peak activations of the adductor longus being highly variable in the current study. The variability most likely resulted from the speed and dynamic nature of the soccer kicking motion in combination with the use of surface electrodes to collect data. Future work might use fine wire electrodes in order to collect muscle activation data during a movement such as kicking. The between-subject variability in the timing and values of the dependent variables reported in this study are similar to those previously reported in the literature for similar movement tasks (Levanon and Dapena, 1998; Nunome et al., 2002; Kellis and Katis, 2007).

Although acute injuries often affect the myotendinous junction, many of the players with chronic groin pain have no obvious findings of injury near the junction. The chronic condition usually causes pain near the insertion of the muscle on the pelvis. Imaging studies and surgical observation do not suggest a tendinopathy. Rather, abnormal Magnetic Resonance Imaging (MRI) signal and traumatic avulsion is visible at the junction of the pubic bone and calcified cartilage (Garrett, 1996; Schilders et al., 2007; Dimitrakopoulou et al., 2008).

Future studies should investigate the potential differences in risk factors for the development of athletic-related groin pain between men and women (Maffey and Emery, 2007). In addition, future work should attempt to collect prospective data on collegiate soccer players in an attempt to understand the differences in kicking mechanics that could lead to injury development.

Conclusions

In summary, the adductor longus appears to be at greatest risk of injury between 30% and 45% of the swing phase. At this point in the kicking cycle, the adductor longus is both eccentrically most active and stretching most rapidly. The coincident timing of maximum activation, maximum rate of lengthening and maximum hip extension angle may indicate

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that the adductor longus assists in controlling hip extension and initiating flexion. This may also indicate that the risk of an adductor longus injury increases with an increase in peak hip extension angle that can occur during powerful kicks. The initiation of hip flexion appears to reduce the rate of adductor longus lengthening. This knowledge could be applied to muscle injury prevention and rehabilitation programs to aid with treatment of adductor-related groin pain.

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References

Akita, K., Niga, S., Yamato, Y., Muneta, T., and Sato, T. (1999). Anatomic basis of chronic groin pain with special reference to sports hernia. *Surgical & Radiologic Anatomy*, *21* (1), 1–5.

Albers, S., Spritzer, C., Garrett, W., and Meyers, W. (2001). MR findings in athletes with pubalgia. *Skeletal Radiology*, *30* (5), 270–277.

Anderson, K., Strickland, S., and Warren, R. (2001). Hip and groin injuries in athletes. *American Journal of Sports Medicine*, *29* (4), 521–533.

Barfield, W. (1998). The biomechanics of kicking in soccer. *Clinics in Sports Medicine*, *17* (4), 711–728.

Bradshaw, C., and McCrory, P. (1997). Obturator nerve entrapment. *Clinical Journal of Sport Medicine*, *7* (3), 217–219.

Brophy, R., Backus, S., Pansy, B., Lyman, S., and Williams, R. (2007). Lower extremity muscle activation and alignment during the soccer instep and side-foot kicks. *Journal of Orthopaedic and Sports Physical Therapy*, *37* (5), 360–368.

Carhart, M. (2003). *Biomechanical analysis of compensatory stepping: Implications for paraplegics standing via FNS*. Arizona State University Tempe, AZ, USA.

Cetin, C., Sekir, U., Yildiz, Y., Aydin, T., Ors, F., and Kalyon, T. (2004). Chronic groin pain in an amateur soccer player. *British Journal of Sports Medicine*, *38*, 223–224.

Cowan, S., Schache, A., Brukner, P., Bennell, K., Hodges, P., Coburn, P. et al. (2004). Delayed onset of transversus abdominus in long-standing groin pain. *Medicine & Science in Sports & Exercise*, *36* (12), 2040–2045.

Delp, S., Loan, J., Hoy, M., Zajac, F., Topp, E., and Rosen, J. (1990). An interactive graphics-based model of the lower extremity to study orthopaedic surgical procedures. *IEEE Transactions on Biomedical Engineering*, *37* (8), 757–767.

DeProft, E., Clarys, J., Bollens, E., Cabri, J., and Dufour, W. (1988). Muscle activity in the soccer kick. In T. Reilly, A. Lees, K. Davids, and W.J. Murphy (Eds.), *Science and football* (pp. 434–440). London: E&F Spon.

Dimitrakopoulou, A., Schilders, E., Talbot, J., and Bismil, Q. (2008). Acute avulsion of the fibrocartilage origin of the adductor longus in professional soccer players: A report of two cases. *Clinical Journal of Sports Medicine*, *18* (2), 167–169.

Dorge, H., Bull-Andersen, T., Sorensen, H., Simonsen, E., Aagaard, H., Poulsen, P. et al. (1999). EMG activity of the iliopsoas muscle and leg kinetics during the soccer place kick. *Scandanavian Journal of Medicine and Science in Sports*, *9*, 155–200.

Dorge, H., Anderson, T., Sorensen, H., and Simonsen, E. (2002). Biomechanical differences in soccer kicking with the preferred and the non-preferred leg. *Journal of Sports Sciences*, *20* (4), 293–299.

Dornan, P. (1971). A report on 140 hamstring injuries. *Australian Journal of Sports Medicine*, *4*, 30–36.

Ekberg, O., Persson, N., Abrahamsson, P., Westlin, N., and Lilja, B. (1988). Longstanding groin pain in athletes. A multidisciplinary approach. *Sports Medicine*, *6* (1), 56–61.

FIFA. Retrieved February 9, 2001, from http://www.fifa.com/mm/document/fifafacts/bcoffsurv/emaga_9384_10704.pdf

Garrett, W. (1990). Muscle strain injuries: Clinical and basic aspects. *Medicine & Science in Sports & Exercise*, *22*, 436–443.

Garrett, W. (1996). Muscle strain injuries. *The American Journal of Sports Medicine*, *24* (6 Supp.), S2–S8.

Gilmore, J. (1998). Groin pain in the soccer athlete: Fact, fiction, and treatment. *Clinics in Sports Medicine*, *17*, 787–793.

Giza, E., Fuller, C., Junge, A., and Dvorak, J. (2003). Mechanisms of foot and ankle injuries in soccer. *The American Journal of Sports Medicine*, *31*, 550–554.

Glick, J. (1980). Muscle strains: Prevention and treatment. *Physician Sportsmed*, *8* (11), 73–77.

Kellis, E., and Baltzopoulos, V. (1995). Isokinetic eccentric exercise. *Sports Medicine*, *19*, 202–222.

Kellis, E., Katis, A., and Gissis, I. (2004). Knee biomechanics of the support leg in soccer kicks from three angles of approach. *Medicine & Science in Sports & Exercise*, *36* (6), 1017–1028.

Kellis, E., and Katis, A. (2007). Biomechanical characteristics and determinants of instep soccer kick. *Journal of Sports Science and Medicine*, *6*, 154–165.

Konrad, P. (2005). *The ABC of EMG: A practical introduction to kinesiological electromyography*. Scottsdale, AZ: Noraxon.

Kuljala, U., Orava, S., and Jarvinen, M. (1997). Hamstring injuries: Current trends in treatment and prevention. *Sports Medicine*, *23*, 397–404.

Lees, A., and Nolan, L. (1998). The biomechanics of soccer: A review. *Journal of Sports Sciences*, *16* (3), 211–234.

Levanon, J., and Dapena, J. (1998). Comparison of the kinematics of the full-instep and pass kicks in soccer. *Medicine & Science in Sports & Exercise*, *30* (6), 917–927.

Lewis, C., Sahrmann, S., and Moran, D. (2009). Effect of position and alteration in synergist muscle force contribution on hip forces when performing hip strengthening exercises. *Clinical Biomechanics*, *24* (1), 35–42.

Lynch, S., and Renstrom, P. (1999). Groin injuries in sport: Treatment strategies. *Sports Medicine*, *28* (2), 137–144.

Maffey, L., and Emery, C. (2007). What are the risk factors for groin strain injury in sport? A systematic review of the literature. *Sports Medicine*, *37* (10), 881–894.

Manolopoulos, E., Papadopoulos, C., and Kellis, E. (2006). Effects of combined strength and kick coordination training on soccer kick biomechanics in amateur players. *Scandinavian Journal of Medicine and Science in Sports*, *16*, 102–110.

Meyers, W., Foley, D., Garrett, W., Lohnes, J., and Mandlebaum, B. (2000). Management of severe lower abdominal or inguinal pain in high-performance athletes. *The American Journal of Sports Medicine*, *28* (1), 2–10.

Morelli, V., and Smith, V. (2001). Groin injuries in athletes. *American Family Physician*, *64* (8), 1405–1414.

Nicholas, S., and Tyler, T. (2002). Adductor muscle strains in sport. *Sports Medicine*, *32* (5), 339–344.

Nunome, H., Asai, T., Ikegami, Y., and Sakuri, S. (2002). Three-dimensional kinetic analysis of side-foot and instep soccer kicks. *Medicine and Science in Sports Exercise*, *34* (12), 2028–2036.

Nunome, H., Lake, M., Georgakis, A., and Stergioulas, L. (2006). Impact phase kinematics of instep kicking in soccer. *Journal of Sports Sciences*, *24* (1), 11–22.

Robinson, P., Barron, D., Parsons, W., Grainger, A., Schilders, E., and O'Connor, P. (2004). Adductor-related groin pain in athletes: Correlation of MR findings with clinical findings. *Skeletal Radiology*, *33*, 451–547.

Schilders, E., Bismil, Q., Robinson, P., O'Connor, P., Gibbon, W., and Talbot, J. (2007). Adductor-related groin pain in competitive athletes. Role of adductor enthesis, magnetic resonance imaging, and entheseal pubic cleft injections. *Journal of Bone and Joint Surgery (Am)*, *89* (10), 2173–2178.

Slavotinek, J., Verrall, G., Fon, G., and Sage, M. (2005). Groin pain in footballers: The association between preseason clinical and pubic bone magnetic resonance imaging findings and athlete outcome. *American Journal of Sports Medicine*, *33* (6), 894–899.

Stanton, P., and Purdam, C. (1989). Hamstring injuries in sprinting: The role of eccentric exercises. *Journal of Orthopaedic & Sports Physical Therapy*, *10*, 343–349.

Stauber, W. (1989). Eccentric action of muscles: Physiology, injury and adaptation. *Exercise & Sports Science Reviews*, *17*, 157–185.

Susmallian, S., Ezri, T., Elis, M., Warters, R., Charuzi, I., and Muggia-Sullam, M. (2004). Laparoscopic repair of "sportsman's hernia" in soccer players as treatment of chronic inguinal pain. *Medical Science Monitor*, *10* (2), CR52–CR54.

Taylor, D., Meyers, W., Moylan, J., Lohnes, J., Bassett, F., and Garrett, W. (1991). Abdominal musculature abnormalities as a cause of groin pain in athletes. Inguinal hernias and pubalgia. *American Journal of Sports Medicine*, *19* (3), 239–242.

van Ingen Schenau, G., Boots, P., DeGroot, G., Snackers, R., and van Woensel, W. (1992). The constrained control of force and position in multi-joint movements. *Neuroscience*, *46* (1), 197–207.

Verrall, G., Slavotinek, J., and Fon, G. (2001). Incidence of pubic bone marrow oedema in Australian rules football players: Relation to groin pain. *British Journal of Sports Medicine*, *35* (1), 28–33.

Verrall, G., Hamilton, I., Slavotinek, J., Oakeshott, R., Spriggins, A., Barnes, P. et al. (2005). Hip joint range of motion reduction in sports-related chronic groin injury diagnosed as pubic bone stress injury. *Journal of Science & Medicine in Sport*, *8* (1), 77–84.

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Verrall, G., Slavotinek, J., Fon, G., and Barnes, P. (2007). Outcome of conservative management of athletic chronic groin injury diagnosed as pubic bone stress injury. *American Journal of Sports Medicine*, *35* (3), 467–474.

Winter, E., and Brookes, F. (1991). Electromechanical response times and muscle elasticity in men and women. *European Journal of Applied Physiology and Occupational Physiology*, *63*, 124–128.

Wong, P., and Hong, Y. (2005). Soccer injury in the lower extremities. *British Journal of Sports Medicine*, *39*, 473–482.

Zarins, B., and Ciullo, J. (1983). Acute muscle and tendon injuries in athletes. *Clinics in Sports Medicine*, *2*, 167–182.

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